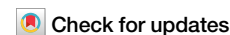




Leveraging the wearable 1-lead ECG signal. From cardiac rhythm to cardiac function assessment



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Traditional 12-lead ECGs offer comprehensive insights into the electrical activity of the heart, but require clinical settings and expert interpretation, which limits their accessibility. Smartwatch 1-lead ECGs can be recorded at home, allowing more frequent and rapid monitoring, opening opportunities for early adverse event detection and enhanced patient autonomy. This study investigates whether 1-lead ECGs can provide clinically meaningful information beyond heart rhythm assessment. Using explainable deep learning models, we predict left ventricular function (LVF) from both 1-lead and 12-lead ECGs in a post-myocardial infarction population, and compare their respective performances. Our findings demonstrate that LVF can be accurately predicted from 1-lead ECGs alone (AUC = 0.883), nearly matching the predictive performance of 12-lead ECGs (AUC = 0.897). Explainability analyses further reveal that the models leverage physiologically plausible ECG features, supporting the validity of this approach. These results suggest that 1-lead ECGs, when combined with explainable AI, have the potential to support broader clinical applications and empower patients, particularly in resource-limited or remote settings where access to traditional cardiac diagnostics remains constrained.

Cardiovascular disease remains the leading global cause of death, underscoring the need for accessible and accurate diagnostic tools. The 12-lead electrocardiogram (ECG) is the clinical gold standard for assessing cardiac electrical activity and diagnosing a wide range of heart conditions. However, its application is largely confined to clinical settings due to the need for specialized equipment and trained personnel.

The 1-lead ECG offers a portable and cost-effective alternative. Devices such as smartwatches and chest patches enable patients to record ECGs independently, facilitating continuous and remote monitoring. However, the reduced spatial resolution of 1-lead ECGs limits their diagnostic scope, particularly for detecting structural abnormalities. These devices have primarily been optimized for rhythm analysis, especially atrial fibrillation (AF) detection, with several FDA- and EMA-approved systems (e.g., Apple Watch and KardiaMobile) achieving high accuracy in real-world applications^{1–3}.

Deep learning has significantly advanced the interpretability and accuracy of ECG-based diagnostics^{4,5}. Explainable AI methods have enabled more transparent and trustworthy models for AF and other arrhythmias^{6–8}, supporting broader adoption in clinical practice.

Beyond arrhythmias, recent work has extended 1-lead ECG applications to detecting heart failure⁹, chronic kidney disease¹⁰, and age-related cardiac changes¹¹. Moreover, predictive modeling using individual leads,

such as Lead I or II, has shown potential for detecting myocardial infarction, left ventricular dysfunction, and myocardial injury, although typically with reduced accuracy compared to full 12-lead recordings^{12–15}.

Despite these advances, few studies have directly compared wrist-based 1-lead ECGs with full 12-lead ECGs acquired from the same patients. Most prior work has either focused on specific conditions (e.g., arrhythmias) or used leads derived from 12-lead datasets, limiting their applicability to real-world wearable recordings. Moreover, while arrhythmia detection has dominated the use of 1-lead ECGs, their potential to identify structural cardiac changes, such as ventricular dysfunction, remains underexplored, although such abnormalities often leave subtle but detectable signatures in the electrical signal. Attia et al. (2022) demonstrated that left ventricular dysfunction (LVD) could be predicted from a 1-lead smartwatch ECG with reasonable accuracy (AUC = 0.885) in a general cardiac patient population¹⁶. However, to our knowledge, left ventricular function (LVF) assessment from 1-lead smartwatch ECGs has not yet been evaluated with explainable AI, nor in the post-myocardial infarction setting, a population in which early detection of ventricular dysfunction is particularly relevant. The post-myocardial infarction population is homogeneous, since scar tissue is the main cause of abnormalities. The confined heterogeneity of this population is driven primarily by the location and severity of myocardial scar. This is an important aspect that can be leveraged in explainable AI models.

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Table 1 | Comparison of all models for both reconstruction (MSE) and prediction of LVF (AUROC, AP, and HBC)

Model	MSE↓	AUROC ↑	AUROC(ovr) ↑	AP ↑	HBC ↑
12-lead	0.027	0.897	0.773	0.600	0.991
	[0.010–0.087]	[0.887–0.906]	[0.768–0.778]	[0.573–0.626]	[0.857–1.000]
12-lead	0.007	0.805	0.700	0.357	0.980
(Lead I)	[0.002–0.036]	[0.792–0.816]	[0.693–0.706]	[0.332–0.382]	[0.667–1.000]
1-lead	0.004	0.883	0.718	0.511	0.959
(pretrained)	[0.001–0.011]	[0.881–0.886]	[0.715–0.720]	[0.501–0.521]	[0.632–1.000]
1-lead	0.004	0.863	0.719	0.459	0.961
	[0.002–0.012]	[0.859–0.866]	[0.717–0.721]	[0.449–0.469]	[0.647–1.000]

The AUROC, AP, and HBC scores are shown for the prediction of the moderate/severe impaired (positive) vs. normal/mild impaired groups; the AUROC (ovr) score shows the averaged one-versus-all AUROC for all four strata. The AUROCs, AP, and HBC metrics show the bootstrapped mean (and 95% confidence interval) on the test set.

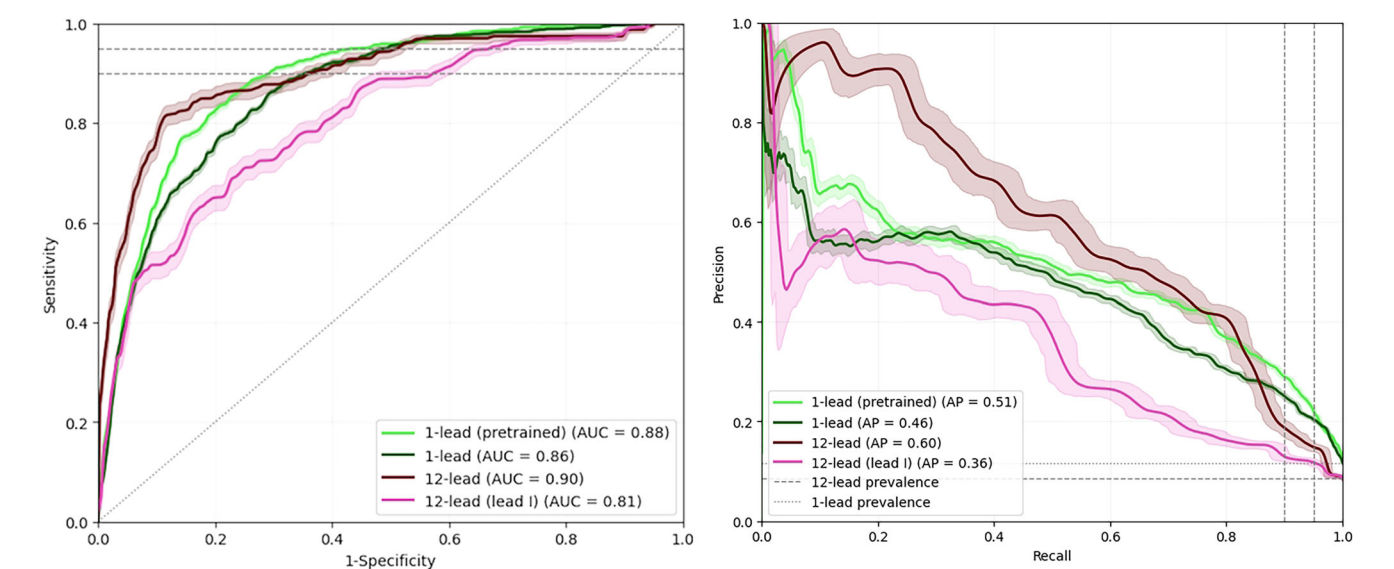


Fig. 1 | ROC and PR analysis, including 95% confidence intervals, comparing the four models. Gray vertical lines indicate recall/sensitivity of 0.9 and 0.95. (left) The ROC curves. (right) The PR curves. AP average precision.

This study conducts a head-to-head comparison of wrist-based 1-lead ECGs, recorded at home as part of the follow-up protocol, and standard 12-lead ECGs, matched by patient and acquisition time, in individuals recovering from myocardial infarction. Focusing on cardiac morphology, we apply deep learning and explainable AI to evaluate whether wearable 1-lead ECGs can move beyond rhythm analysis and provide clinically meaningful insight into structural cardiac abnormalities.

Results
LVF prediction

In total, 7,118 1-lead ECG signals and 7,593 12-lead ECG signals were labeled with LVF. Respectively, 14% and 11% of these were labeled with severe/moderate impaired LVF.

Several data quality issues were identified in the 1-lead recordings, including devices worn on the wrong wrist, leading to signal inversion, device sharing among multiple individuals, and acquisition during non-resting states. These factors introduce substantial noise and may impair model performance. To mitigate this, we applied a correlation-based sieve that utilized preceding and subsequent signals from the same patient to exclude low-quality recordings that showed low correlation with the other signals. After sieving, 5,480 1-lead signals (1,638 excluded) comprising 108,364 heartbeats from 460 patients (77.6% male, age 61.4 ± 11.4 years) were retained. For the 12-lead recordings, no comparable quality concerns

were observed; 7,197 signals (396 excluded) and 19,933 heartbeats from 1,856 patients (73.4% male, age 62.7 ± 11.9 years) were included. The test sets consisted of 900 1-lead signals and 559 12-lead signals from 144 patients.

Table 1 and Fig. 1 summarize the performance of the four models for LVF prediction on the test set. As anticipated, the full 12-lead model achieved the highest predictive performance across all metrics. The 1-lead models demonstrated reasonable discriminative ability, with only a modest reduction in performance compared to the 12-lead model. Pretraining the 1-lead model using lead I of the 12-lead ECG improved AUC and AP but did not improve HBC and AUROC(ovr). The lead I model exhibited the lowest AUC yet ranked second HBC after the full 12-lead model. Supplementary Table 1 (see Supplementary Information) presents the performance of the model divided by sex. Across all models, AUROC is consistently higher in the male subset, though the remaining metrics do not allow a definitive conclusion regarding sex-related performance differences.

Operating points

In Fig. 1, sensitivity at 0.9 and 0.95 are indicated with grey dotted lines. At sensitivity 0.90, specificities were: 0.64 [0.58–0.69] (12-lead), 0.43 [0.39–0.55] (lead I), 0.71 [0.70–0.72] (pretrained 1-lead), and 0.65 [0.64–0.66] (1-lead). All models showed high intra-recording stability, as reflected by elevated HBC scores: most QRS complexes within a single recording were consistently assigned to the same binary class. This

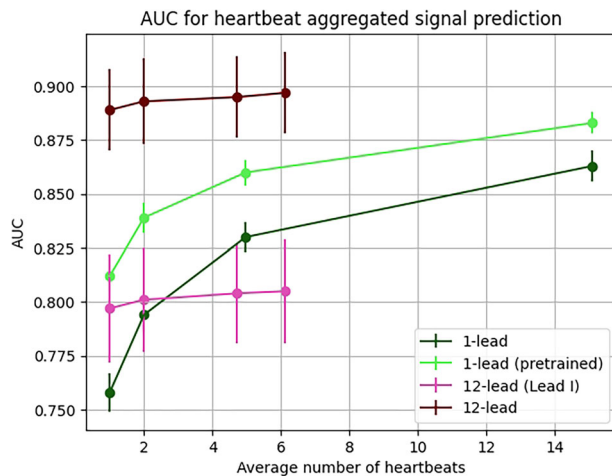


Fig. 2 | AUROC values for intra-signal aggregation of predictions. 1-lead models benefit significantly more from post-processing than 12-lead models.

robustness underscores the reliability of signal-level predictions. Figure 2 illustrates the AUC when the predictions at the heartbeat level were aggregated by averaging within each signal.

Interpretability

Figures 3, 4 show attribute manipulation of a heartbeat of the same patient with the pre-trained 1-lead and the full 12-lead model, respectively. The attribute manipulation can be interpreted as the rationale behind the model's prediction. The 12-lead model shows a much more diverse set of features in multiple leads. Quantitative assessment of these features after myocardial infarction is currently not part of the clinical routine.

Discussion

This study provides a comparison of LVF prediction using matched 1-lead and 12-lead ECG datasets. As expected, the 12-lead model achieved the highest overall performance; however, the performance gap with the 1-lead models was modest, indicating that a substantial portion of LVF-relevant information present in the full 12-lead ECG is also captured in a single lead. Model performance was comparable to, and in some instances slightly exceeded, previously reported results in broader cardiac populations^{7,16} (AUCs 0.83 and 0.89, respectively). Since early detection of ventricular dysfunction is particularly relevant in our post-myocardial infarction population, the findings of our study bear significant importance.

At a sensitivity of 0.90, specificity scores were 0.64 [0.58–0.69] (full 12-lead), 0.43 [0.39–0.55] (lead I), 0.71 [0.70–0.72] (pretrained 1-lead), and 0.65 [0.64–0.66] (1-lead). These operating points are usable in clinical pathways. The scores show the percentage of healthy patients that should not be reviewed for a test that correctly predicts left ventricular dysfunction with an accuracy of 0.9. They could be used, for example, to trigger manual review or flag patients as likely unchanged/low risk, especially in settings where 12-lead acquisition is not feasible. Notably, averaging heartbeat-level predictions within a recording preferentially improved the 1-lead models, consistent with (i) the larger number of heartbeats available in the longer 1-lead signal (30 s vs. 10 s) and (ii) noise cancellation through aggregation. Together with the observed high heartbeat consistency, this also supports the benefit of temporal aggregation over multiple recordings.

A distinctive aspect of this work is the use of attribute manipulation to provide a transparent view of model reasoning. The 12-lead model leveraged a broader, multi-lead feature set (e.g., R-peak amplitude, notched R-peaks, T-wave morphology, see Fig. 4), whereas the 1-lead models used features that resembled the 12-lead feature set for lead I, which restricts interpretability and model performance, see Fig. 3. This raises important hypotheses: (i) whether 1-lead LVF prediction is most effective in subtypes with global or lateral abnormalities visible on lead I; (ii) whether multi-lead configurations

are essential for anterior or inferior infarctions; and (iii) how explainability can guide lead selection for wearable or resource-limited settings. The observed performance drop for the lead I model compared to the full 12-lead model reinforces the value of complementary spatial information, yet the improvement of the 1-lead model after heartbeat aggregation suggests that temporal averaging may partially compensate for reduced spatial coverage. Future work should validate these hypotheses and explore whether explainability-driven insights can inform adaptive acquisition protocols or subtype-specific prediction models.

While wrist-based 1-lead ECGs have primarily been optimized for rhythm analysis, such as atrial fibrillation detection, this study demonstrates their potential for structural and functional cardiac assessment. Although 1-lead ECGs are not intended to replace 12-lead ECGs, the combination of reasonable discriminative performance, transparent mechanism, and aggregation-driven stability supports 1-lead ECGs as a practical tool for screening and potentially longitudinal monitoring. It could be used as a scalable clinical complement for (automated) early warning and home follow-up where high-quality 12-lead acquisition is not possible.

We addressed 1-lead data quality challenges (device orientation errors, device sharing, non-rest acquisition) using an autocorrelation-based filter informed by adjacent recordings from the same patient. However, differentiating true pathological inversions from orientation-related inversions remains non-trivial, since several conditions transiently invert peaks or alter large ECG segments. We explored heartbeat autocorrelation, landmark-based checks, and learned classifiers with mixed success. Future research directions to address these issues and extend currently methodology include: (i) acquisition protocol refinement to reduce incorrect device orientation and activity artifacts, (ii) robust automated quality control that combines signal-to-signal and even beat-to-beat consistency and anomaly detection, (iii) temporal modeling (e.g., multi visit aggregation) to convert stable beat-level consistency into patient-level trajectories, (iv) model performance analysis in clinical subtypes (global/lateral repolarization abnormalities vs. regional abnormalities), (v) clinical validation with calibrated, use case specific thresholds (e.g., fixed 0.90 specificity), including health system impact metrics, and (vi) reformulating LVF prediction as a regression rather than a classification task. Although LVF is represented by four discrete classes in the current study, while clinically practical and more widely available, it is derived from an underlying continuous measurement. Training models on a regression objective would preserve this continuity and improve the sensitivity of longitudinal change detection: subtle shifts near class boundaries would no longer be obscured by label discretization, and large changes confined within a single class would no longer go undetected.

The VAE models trained in this study are likely not directly generalizable to broader cardiac populations, as the explainability framework deliberately leverages the confined and well-characterized heterogeneity inherent to the post-myocardial infarction population. This is an intentional design choice: by restricting the training distribution, the latent space of the VAE can be shaped to encode clinically meaningful variation, allowing more interpretable disentanglement of pathophysiological features. Extending this approach to more heterogeneous cardiac populations, encompassing, for example, patients with heart failure or LVD due to nonischemic etiologies, would require substantially larger and more diverse training sets, and may introduce confounding sources of variability that complicate the model explainability. Although the absence of external validation represents a limitation of this study, the primary aim of this exploratory work was to compare the clinical utility of morphological monitoring using the 12-lead versus the 1-lead ECG.

The 12-lead ECG remains the gold standard for LVF prediction, yet 1-lead models achieve clinically meaningful performance, particularly when predictions are aggregated. Explainability highlights a confined single-lead feature set in the 1-lead models versus multi-feature, multi-lead reasoning in the 12-lead model.

This study demonstrates that LVF can be assessed from wearable 1-lead ECGs with clinically meaningful accuracy (AUC = 0.883),

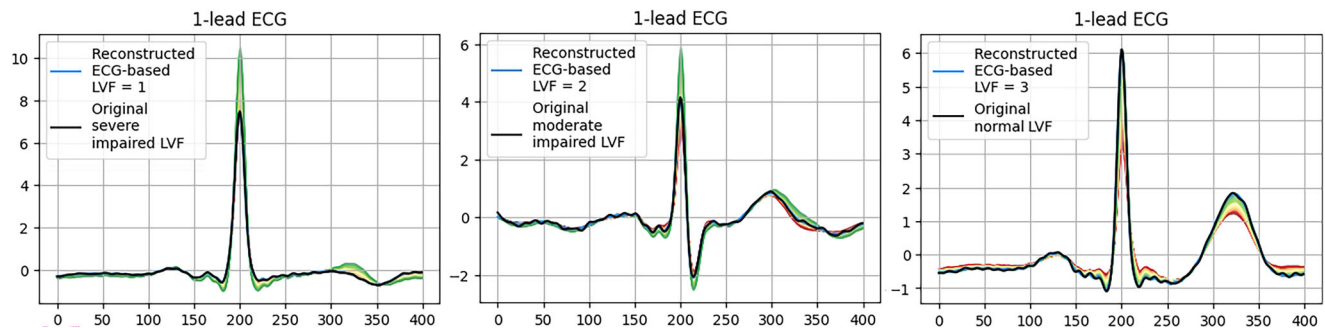


Fig. 3 | Attribute manipulation in three heartbeats with different LVF impairment. The black line shows the origin heartbeat of the 1-lead ECG; the blue line (largely covered by the black line here) shows the reconstructed heartbeat by the model. The green to red distributions around the reconstructed signal are the result of attribute manipulation and can be interpreted as the reason that the model

classified the signal with the LVF value shown in the legend (0 = severely impaired, 1 = moderately impaired, 2 = mildly impaired, 3 = normal). The green signal shows the 1-lead signal that would be classified as having a normal LVF. The green signal in all figures shows an R-peak with a larger amplitude, more pronounced Q- and S-waves and a T-wave with a slightly larger area.

approaching the performance of standard 12-lead ECGs (AUC = 0.897). Combined with explainable AI, these tools highlight the specific waveform features, R-peak amplitude, T-wave morphology, and Q- and S-wave depth that drive predictions, providing mechanistic transparency to support clinical decision-making. At a sensitivity of 0.90, the pretrained 1-lead model achieved a specificity of 0.71, suggesting utility as a screening tool to flag patients at elevated risk of ventricular dysfunction between scheduled clinical visits. This is particularly relevant in post-MI follow-up, where early detection of deteriorating function carries significant prognostic implications. Several barriers must be addressed before these models can be deployed in routine clinical care. First, prospective validation in independent post-MI cohorts and broader cardiac populations is needed to confirm generalizability beyond the single-center dataset used here. Second, robust automated quality control pipelines must be developed to handle real-world recording artifacts prevalent in unsupervised home monitoring. Third, clinical workflow integration requires careful threshold calibration to specific use cases, and health system impact metrics should be evaluated. Finally, regulatory pathways for AI-based cardiac function monitoring from consumer wearables remain nascent, and prospective studies demonstrating clinical outcome benefit will be essential for adoption.

Methods

Data

The study uses matched 1-lead and 12-lead ECG datasets from the same cohort, see Fig. 5. All data were obtained from patients diagnosed with acute coronary syndrome, between 2018 and 2023 and treated at the Leiden University Medical Center, Netherlands. The 1-lead dataset contains 63,950 raw 30-s, 1-lead ECG signals (300 Hz) from 708 patients (77.3% male, age 59.5 ± 10.8 years), recorded at home using a Withings ECG Smartwatch as part of a clinical telemonitoring program^{17,18}. The 12-lead dataset contains 77,005 raw 10-s, 12-lead ECG signals (500 Hz), from 6,703 patients (76.3% male, age 60.5 ± 11.3 years). The study protocol, identified as nWMO-DIV2_2022006, received institutional review board approval (Niet-Wet medische wetenschappelijk onderzoek (nWMO) commissie Divisie 2), which waived the requirement for informed consent for the use of pseudonymized ECG data, since it is considered infeasible to obtain informed consent in retrospect for such a large number of patients.

Data preprocessing

Preprocessing of both 1-lead and 12-lead ECG signals followed the procedure described in Van der Valk et al.⁷, see Fig. 6 for an overview. The raw ECG signals were segmented into individual heartbeats, each defined as an 800 ms interval centered on the R-peak, 400 ms preceding and 400 ms following. This interval is chosen to balance under- and over-segmentation and thereby either missing parts of the heartbeat or including previous or subsequent heartbeats. R-peak detection was performed using a method

inspired by RPNNet, a U-Net-structured CNN, previously trained on manually annotated peak locations as described in Vijayarangan et al.¹⁹.

Because the 1-lead recordings were 30 s in duration compared to the 10 s duration of the 12-lead recordings, the 1-lead data yielded a greater number of segmented beats. Given the low intra-recording variability and to maximize the available training data for the 1-lead model, all valid heartbeats were retained for this modality. Segmented beats containing excessive noise or flatline segments were excluded; for the 12-lead case, exclusion was applied if any lead exhibited such artifacts. For details on the exclusion procedure, see Van der Valk et al.⁷.

Left ventricular function

We selected LVF as a prediction target due to its reflection of structural changes after MI and the presence of well-characterized ECG features. Several of these, such as the first QRS integral, ST integral, ST-T integral, and ST-T area (integral of the absolute values), have shown discrimination in recovering MI patients. The predictive capacity of these features differed per lead and MI location^{20,21}. T-wave morphology changes, including inversion and subsequent reinversion, have also been linked with favorable LVF recovery²².

ECG signals were labeled with four ordinal LVF categories: normal function, mild impairment, moderate impairment, and severe impairment (LVEF $\geq 55\%$, 45%–54%, 35%–44%, and $<35\%$, respectively). Labeling was performed using a large language model (LLM)-based classification of echocardiogram reports obtained as part of the routine clinical workflow. This method is validated and described by Van der Loo et al.²³. An echocardiogram report was used as a label if the examination was conducted within 7 days before or after the corresponding ECG recording. A 1 day instead of a 3 day margin was applied when the ECG was obtained within two weeks after cardiac intervention, as the LVF can still change considerably in the first weeks after cardiac intervention. The reports themselves were based on visual interpretation of the echocardiograms by clinicians.

Model overview and training

LVF prediction from both 1-lead and 12-lead ECGs employed the same model and (pre-)training strategy as successfully employed in Van der Valk et al.⁷. A partial β -variational autoencoder (VAE) (Fig. 6) was used for both modalities. The architecture consisted of seven recurrent convolutional blocks in both the encoder and decoder. Four separate model instances were trained: (i) a 12-lead model, a model trained and evaluated on the 12-lead ECGs, (ii) a lead I model, a model trained and evaluated on lead I of the 12-lead ECGs, (iii) a pretrained 1-lead model, a model pretrained on lead I of the 12-lead ECG and finetuned (retrained using the same training parameters, no frozen layers) and evaluated on the 1-lead ECGs, and (iv) a 1-lead model, a model trained and evaluated on the 1-lead ECGs. The models expect an input of shape 12×400 ; therefore, for the 1-lead recordings, each

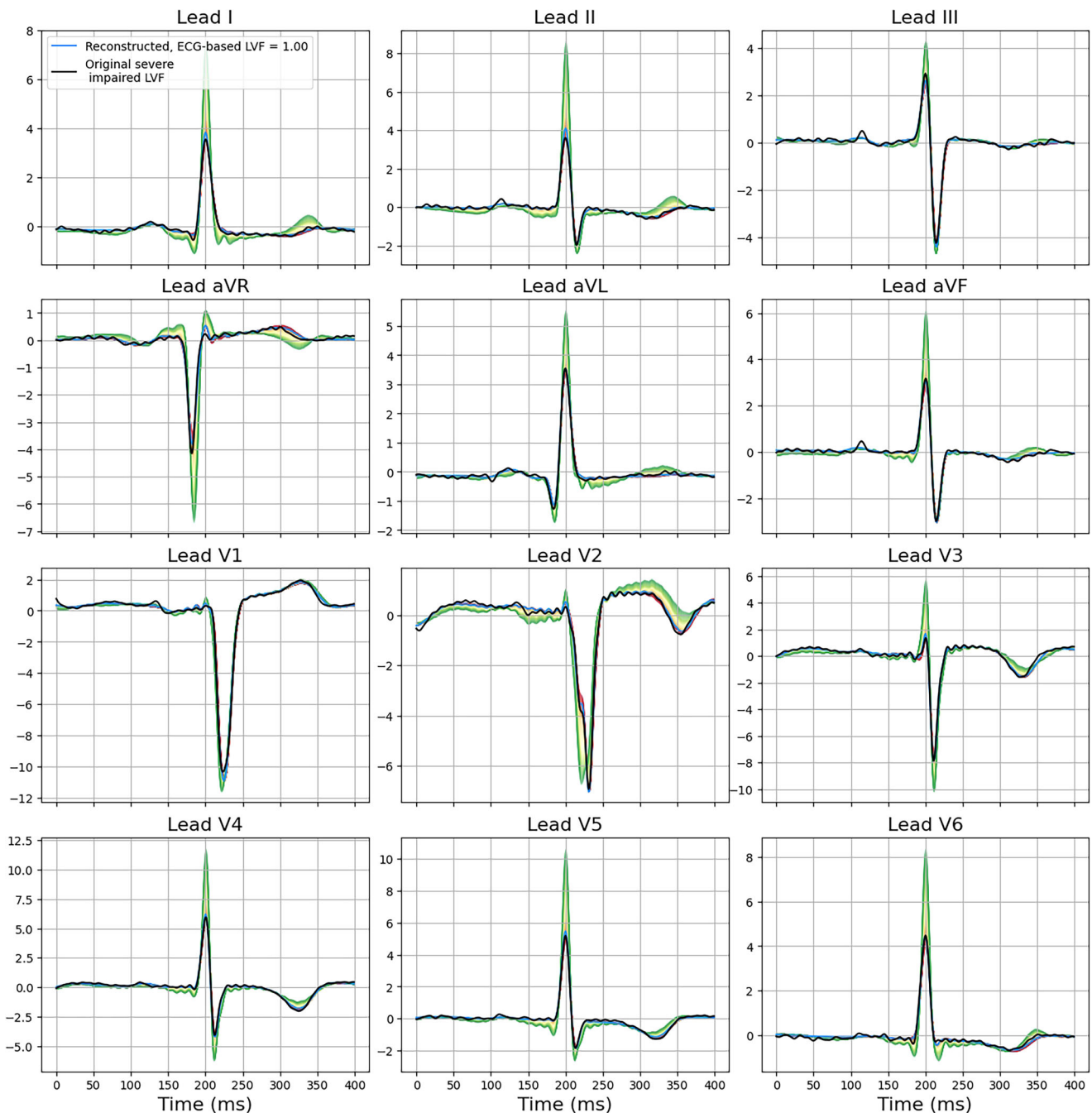


Fig. 4 | Attribute manipulation in a heartbeat with severe LVF impairment. The patient and visualization method are the same as in the first plot in Fig. 3, see that caption for detailed explanation. In this case the green signal shows among others a

larger R-peak amplitude, a T-wave with larger area and a R-peak without a notch in lead aVR, V2, aVF, V5, and V6.

1 × 400 heartbeat was duplicated along the lead dimension to match the required input shape.

Feature visualization

The model output and interpretability are shown by means of attribute manipulation as explained in Van der Valk et al.⁷, see Fig. 6. Each ECG heartbeat was encoded into z_{nl} s and z_{ls} , of dimension 599 and one, respectively. Then z was decoded to reconstruct the signal. For attribute manipulation six hypothetical signals were created by replacing z_{ls} with six values: four means for the four LVF groups (S_1 – S_4) and two extremes (S_0 : mean minus one standard deviation of severely impaired group; S_5 : mean plus one standard deviation of normal LVF group). These six patient-specific hypothetical signals were used to illustrate how ECG morphology changes

across LVF impairment levels and which features the model leverages for classification. These signals create a context for the ECG heartbeat w.r.t. LVF, allowing comparison against the whole database. This facilitates signal interpretation and potentially enables interpretation by less trained medical personnel.

Experiments

Both datasets were split into a training set (89% of signals) and a holdout test set (11% of signals). The test set was constructed exclusively from the overlapping portion of the datasets, where both patients and acquisition dates (within a seven-day window) were overlapping, to enable a head-to-head comparison. To prevent overfitting, none of the patients in the 12-lead and the 1-lead test sets were used in any of the training sets. A fivefold cross-

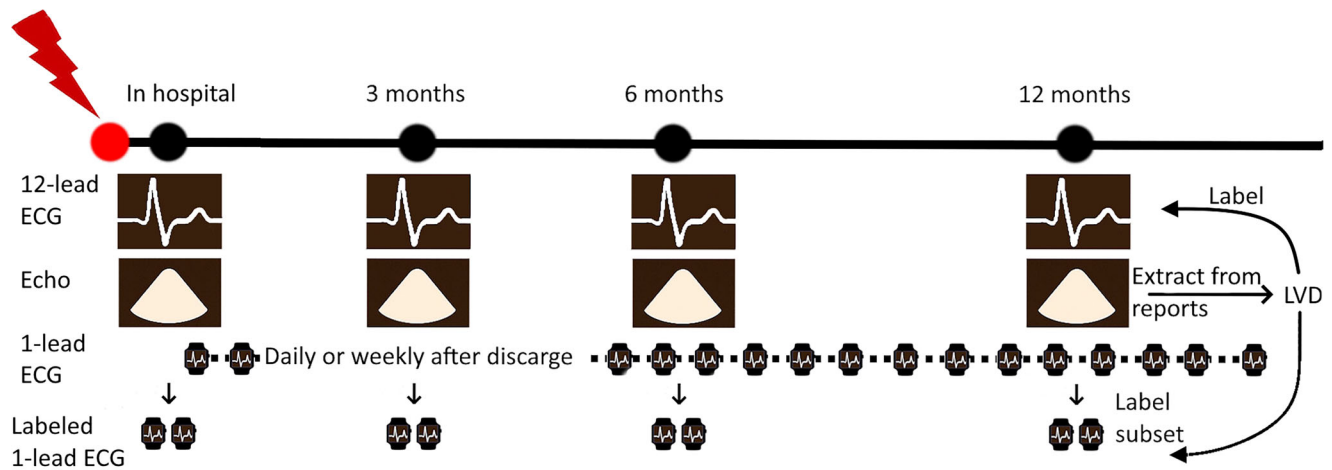


Fig. 5 | Clinical monitoring and data collection. Timeline indicates the 12-lead ECG, 1-lead ECG, and echocardiogram data collection following the clinical protocol. 12-lead ECGs and echocardiograms are collected during hospital visits as part of the clinical routine. 1-lead ECGs are collected at home by patients on suggested

intervals after discharge. A subset of the 1-lead ECGs, when acquisition overlaps with echocardiography, is labeled with the LVF label extracted from the echocardiography reports.

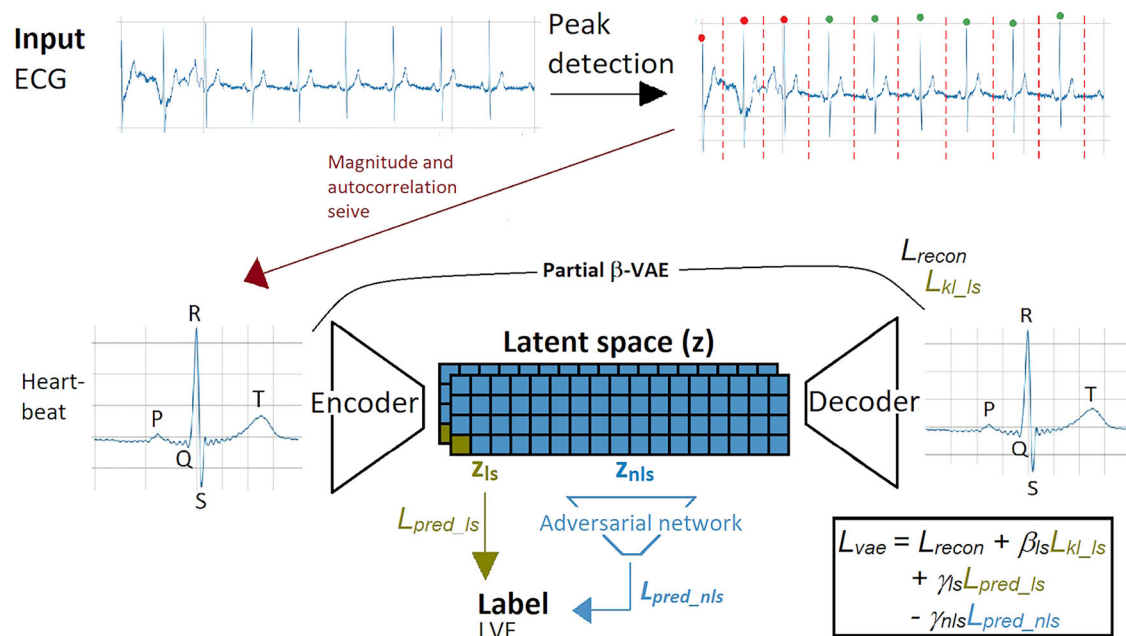


Fig. 6 | Overview of our processing and prediction framework. Initially, the 12-lead ECGs are split into individual heartbeats using RPN¹⁹. Subsequently, the heartbeats are sieved in two stages: the first based on individual magnitude, identifying low magnitude as indicative of a flat signal, and the second utilizing autocorrelation to assess correlation with other heartbeats from the same signal. The remaining heartbeats (indicated with green dots) are used as input for the VAE. The

encoded representation z is derived, comprising z_{nls} and z_{ls} . While z_{ls} directly contributes to LVF label prediction with loss L_{pred_ls} , z_{nls} serves as input for an adversarial multi-layer perceptron (MLP) also predicting the LVF label, introducing loss L_{pred_nls} . The decoder's input is sampled from z to produce the reconstructed heartbeat, subsequently compared with the original input heartbeat, resulting in the reconstruction loss L_{recon} .

validation was performed in both training sets, maintaining an 85:15 ratio between the training and validation sets within each fold. All data splits were done at the patient level and stratified by label when applicable. Training continued until convergence, defined as no improvement in validation loss for 15 consecutive epochs. To mitigate overfitting, we employed balanced sampling, early stopping, L2 regularization on the fully connected layer, and weight decay with the Adam optimizer. To prevent gradient explosion, He initialization was used²⁴, and to avoid numerical overflow, the standard deviations of the posterior $q(z|X)$ were clipped to the interval $[-10, 3]$ prior to sampling. The models were implemented and trained on a 24Gb GPU. Hyperparameter tuning resulted in $\beta_{ls} = 0$, $\beta_{nls} = 0.001$ and $\gamma = 2.0$ for both models. This is in line with the experiments shown by Van der Valk et al.⁷.

Feature evaluation

LVF predictions were evaluated using three classification metrics: the Area Under the Receiver Operating Characteristic Curve (AUROC), average precision score (AP), and a heartbeat consistency score (HBC). The AP and HBC metrics were applied to a binary classification task (positive class: severe/moderate impairment vs. negative class: mild impairment/normal function), reflecting clinically relevant groupings and enabling comparison with previous studies. The AUROC metric was used to evaluate the one-versus-all prediction for all four classes and to evaluate the binary classification task mentioned before. The HBC score was defined as the portion of heartbeats from the same signal that was classified in the same group. The

AUROC and AP scores were calculated with the averaged predictions from all heartbeats in a signal.

Heartbeat reconstruction performance was assessed using the mean squared error (MSE) between the ECG input and the output signal. For intermodel comparisons of prediction scores and MSE, paired t-tests were performed. The confidence intervals (95%) for AUC, AP, and HBC scores were estimated by bootstrapping with $n = 1000$ samples of the combined output distribution (five models). The confidence interval for MSE was estimated directly with the combined output error distribution.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to strict privacy regulation of the Leiden University Medical Center but are available from the corresponding author on reasonable request.

Code availability

Deep learning analyses were performed using Python 3.10 and PyTorch 2.5.1 (CUDA 11.8) for GPU-accelerated model training and inference. All parameters and custom code developed for this study will be made publicly available on [GitHub](#).

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Author contributions

V.V. wrote the main paper text and prepared all figures. M.S. and R.S. gave multiple rounds of feedback on the figures and text. All authors reviewed the final paper. As supervisors, M.S. and R.S. had a major role in the study design and data collection executed by V.V., as head of the cardiology department, D.A. had a major role in the initial study initiation, data collection, and funding acquisition.

Competing interests

The authors declare no competing interests.

Additional information

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